CHINESE-BRAZILIAN PROTOCOL ON DISTRIBUTION OF *CBERS* PRODUCTS

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The Complementary Protocol to the Framework Agreement Between the Government of the People's Republic of China and the Government of the Federative Republic of Brazil on Cooperation in the Peaceful Application of Outer Space Science And Technology on the Cooperation for the *CBERS* Application System[1] was signed on November 12, 2004, during the visit of the President of China to Brazil. It marks a great change. Both countries become suppliers of remote sensing data instead of only users. It is supplemented by the annexed text on *CBERS* Data Policy and regulates the distribution of the Chinese-Brazilian Earth Resources Satellite (*CBERS*) imagery between them and to third parties.

The regulation applies to the data from *CBERS-2*, which was launched on October 21, 2003, and from *CBERS-2B*, *CBERS-3* and *CBERS-4*[2], which are scheduled to be launched in 2006, 2008 and 2010, respectively.

The *CBERS* Program began in 1986 and its first Protocol was signed by both Governments in 1988, in Beijing. It was also the first agreement signed by two developing countries on high space technology cooperation. Brazil and China confronted and overcame many difficulties to accomplish their goals. That is

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[1] The Complementary Protocol to the Framework Agreement Between the Government of the People's Republic of China and the Government of the Federative Republic of Brazil on Cooperation in the Peaceful Application of Outer Space Science and Technology on the Cooperation for the CBERS Application System, Nov. 12, 2004 [hereinafter Protocol] (on file with authors).

[2] It has been agreed that the *CBERS-4* will be launched from the Brazilian Alcantara Launch Center if the necessary conditions to do so exist at that time.

why its first result took eleven years to happen when the experimental *CBERS-1* was launched on October 14, 1999, from the Chinese base in Taiyuan. Both countries have invested US$ 300 million in *CBERS-1* and *2*: US$ 182 million by China and US$ 118 million by Brazil. *CBERS 2B* will cost US$ 34 million for China and US$ 15 million for Brazil.[3]

The current Protocol is a result of the Memorandum of Understanding signed on May 24, 2004. It extends the scope of the *CBERS* Program to establish criteria for commercializing *CBERS* data. The Chinese National Space Administration (CNSA) and the Brazilian Space Agency (BSA) were designated to implement the agreement.

The Protocol establishes the requirements for tasks, functions, and specifications of the application system's infrastructure; consolidates the technical organization of the application system's infrastructure; and defines a plan for developing and producing an application system that shall privilege Chinese and Brazilian companies. It also creates a cooperation program to develop software for the reception and processing system for *CBERS* data for use by other countries.

The Protocol develops and improves software for *CBERS* data application and for generating products for end users; provides for meetings for exchanging experiences regarding *CBERS* data; and promotes technical exchange of *CBERS* data among users from China, Brazil, and other countries. It also establishes and implements criteria and patterns for evaluating *CBERS* products; and, promotes and summarizes discussions of feedback from users in order to improve *CBERS* sensors and then propose new onboard sensor technical requirements.

In accordance with the Protocol, disputes and doubts regarding its terms will be resolved through mutual consultations between both countries. The questions related to intellectual property will be established in specific agreements which will take into account the national legislation of both parties and international rules accepted by them. The Protocol will remain

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[3] Earth Observation General Coordination Division of INPE (Coordenação-Geral de Observação da Terra) (research on file with authors).

valid for five years and will be renewed automatically for successive periods, unless one party notifies the other six months in advance that it wishes to terminate the agreement.

The *CBERS* Data Policy, which is annexed to the Protocol, establishes the policy for commercializing *CBERS* data and includes provisions for receiving, processing, and disseminating *CBERS* images to countries other than Brazil and China. Accordingly, *CBERS* images will be made available to any country or organization through a network of licensed representatives operating an application system that can receive and process the *CBERS* data.

International ground stations will not have access to the onboard data recorder (OBDR), which will be operated exclusively by the China Center for Resources Satellite Data and Applications (CRESDA) and the Brazilian National Institute for Space Research (INPE). Each ground station will receive raw data only for specific regions as determined by contract and will then process it into image products, which will then be distributed to users. The licensing of *CBERS* data downlinks will be based on fees which are charged on a per-minute basis. Distribution of *CBERS* images to third parties will be done solely on the basis of the international price list, as agreed to by China and Brazil. China and Brazil may, in special cases, upon mutual consultations, decide on the transfer of data free of charge. Mozambique, Angola and some other African countries will be benefited by this exception, receiving data free of charge from Brazil.

Brazil provides 30% of the *CBERS* budget. Nonetheless, it will receive the same revenues as China from the distribution of *CBERS* data. Moreover, ground stations operated by INPE[4] and by CRESDA[5] will have unlimited access to all data collected within their footprint.

There are guidelines that regulate the use of the OBDR. The number of hours available each month for use of the OBDR

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[4] Instituto Nacional de Pesquisas Espaciais, *at* www.inpe.br (last visited Jan. 9, 2006).

[5] China Center for Resource Satellite Data and Applications, *at* www.cresda.com.cn/ (last visited Jan. 9, 2006).

will be established periodically by the engineering teams of *CBERS*. INPE and CRESDA will equally share the available time, for example, 50% to CRESDA and 50% to INPE, on a non-cumulative basis, so unused hours in a month cannot be accumulated for subsequent periods; images stored in the OBDR will be downloaded at the Brazilian and Chinese ground stations. Data downlinks to ground stations will have priority over the use of OBDR, except in the case of emergency situations, as determined by INPE and CRESDA.

The interface between a satellite and a ground station is regarded as the intellectual property of INPE and CRESDA and will not be disclosed to third parties unless agreed to in writing by both parties. INPE and CRESDA will encourage Brazilian and Chinese companies to act as suppliers of components and equipment for *CBERS* application system infrastructure.

As for Chinese and Brazilian ground stations, each party is free to decide on its own development strategy that will preferably be carried out only by national companies. The parties agree that any component of the ground station that cannot be built by its national industry will first be requested from the other partner, before any contracts are placed in the international market.

The document also deals with the licensing policy for international ground stations. The licensed representative will commercialize *CBERS* data downloaded to ground stations based on an annual fixed fee, to be determined by INPE and CRESDA. The annual fee will be determined by the conditions of the ground stations, including geographical location and tracking system footprint. The licensed representatives shall hold annual meetings with their customers and promote the diffusion of any news and decisions regarding *CBERS*.

The document adopts as well terms and conditions for commercial agreements between licensed representatives and *CBERS* data distributors. The images sold within the distributor's national market may not be exported abroad.

Brazil is developing a rich experience regarding satellite data satellite as a public good. *CBERS-2* images have been distributed freely within the country. Since June 2004 to November 2005 more than 150,000 images were distributed in the

Brazilian territory to more than 14,000 users – more than 2,000 institutions. Among these institutions only 5 took 2% of the images and 25 took between 2% and 0.5%; the average of requests for *CBERS* data is about 800 per week; over 2,100 images are downloaded per week; each user takes 10.5 images. 22.9% of the demand for *CBERS* images is from government; 25.9% from educational and research institutions; and 51.2% from the private sector, including companies and individuals.[6]

Thus, Brazil has largely been disseminating the culture of utilization of remote sensing data in a wide variety of different forms, creating a wide and equitable internal market for its products. It used to purchase 20,000 images from *Spot-5* and 12,000 from *Landsat*, and distribute 1,000 *Landsat* images each year.[7] The free distribution data policy will last at least two years.

The *CBERS* Program has become a long term cooperative system. *CBERS-5, 6, 7* and perhaps more can be expected. Brazil and China have signed 15 bilateral agreements, protocols and minutes of understanding since 1982. They consider their collaborative efforts as "strategic partnership".

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[6] Earth Observation General Coordination Division of INPE, *supra* note 3.
[7] *Id.*

COMPLEMENTARY PROTOCOL TO THE FRAMEWORK AGREEMENT BETWEEN THE GOVERNMENT OF THE PEOPLE'S REPUBLIC OF CHINA AND THE GOVERNMENT OF THE FEDERATIVE REPUBLIC OF BRAZIL ON COOPERATION IN THE PEACEFUL APPLICATION OF OUTER SPACE SCIENCE AND TECHNOLOGY ON THE COOPERATION FOR THE *CBERS* APPLICATION SYSTEM

The Government of the People's Republic of China, and The Government of the Federative Republic of Brazil (hereinafter referred to as "The Parties"), For the purpose of further strengthening the cooperation in the field of peaceful use of space technology between the Parties;

In order to further promote the role of space technology in the social, economic and cultural development of the two countries;

Recalling the terms of the Framework Agreement between the Government of the People's Republic of China and the Government of the Federative Republic of Brazil on Cooperation in the Peaceful Applications of Outer Space Science and Technology, signed in Beijing, on November 8th, 1994;

Recalling the terms of the Protocol on Cooperation in Space Technology between the Government of the People's Republic of China and the Government of the, Federative Republic of Brazil signed in Brasilia, on September 21st, 2000;

Considering that the Parties have signed a Protocol for the extension of the *CBERS* program, and have agreed to establish a cooperation project to develop China-Brazil Earth Resources Satellites (*CBERS*) numbers 03 and 04, with provisions for cooperation in the application system; and

Recalling the terms of the Memorandum of Understanding on the Cooperation for the Development of an Application System for the China-Brazil Earth Resources Satellite Program, signed in Beijing on May 24th, 2004,

Have agreed as follows:

ARTICLE I

The Parties agree to extend the scope of the *CBERS* Program by setting up a cooperative framework which will enable

establishing the *CBERS* application system through specific cooperation projects, which shall include the distribution of *CBERS* products to countries other than China and Brazil.

ARTICLE II

All cooperation projects under this Protocol are subject to the general terms and conditions agreed between China and Brazil as regards the *CBERS* Program.

ARTICLE III

The Parties designate respectively the Chinese National Space Administration (CNSA) and the Brazilian Space Agency (AEB) as the entities responsible for implementing the actions agreed in this Protocol and to oversee the implementation of all cooperation projects proposed under this Complementary Protocol.

The CNSA and the AEB may, each on its side and under its discretion, entrust other entities with the responsibilities set forth in this Article.

Major cooperation projects and decisions should be approved by each Party after assessment by both the Joint Project Committee of the *CBERS* Program (hereinafter referred to as JPC) and the Program Coordination Committee between China and Brazil (hereinafter referred to as PCC).

ARTICLE IV

The Parties agree to appoint or to establish a proper organization or joint committee which will be responsible for organizing and implementing the cooperation projects proposed under this Complementary Protocol.

ARTICLE V

The Parties agree to implement, through specific cooperation projects, the following work:

1. Jointly establish the requirements for the tasks, functions and specifications of the Application System Infrastructure.

2. Jointly consolidate the overall and implementation technical schemes of the Application System Infrastructure.

3. Jointly define a development and production plan for the Application System Infrastructure that prioritizes the supply of technology by Chinese and Brazilian companies.

ARTICLE VI

1. The Parties agree to distribute *CBERS* products to countries other than China and Brazil subject to the conditions set in the *CBERS* Data Policy, which is annexed to and as an integral part of the Complementary Protocol.

2. The Parties shall share equally the revenues of the distribution of *CBERS* products, according to the *CBERS* Data Policy.

ARTICLE VII

The Parties agree to establish a specific cooperation project aimed at the construction of data receiving and processing systems for *CBERS* satellites, as part of the effort of distributing *CBERS* products in countries other than China and Brazil. The specific cooperation project shall be submitted to the PCC, through the JPC, for approval by the Parties and shall conform to the *CBERS* Data Policy defined and approved by the Parties.

ARTICLE VIII

The Parties agree to carry out the following activities regarding cooperation and development of *CBERS* data applications:

1. Develop and extend *CBERS* data application software and end users' products.

2. Hold meetings for exchanging experience on *CBERS* data applications.

3. Jointly promote technical training on *CBERS* data applications for users from China, Brazil and other countries.

4. Jointly establish and implement criteria and standards for the evaluation of *CBERS* image products and procedures for image calibration and quality assessment.

5. Jointly compile and promote discussions regarding users' requirements for the sensors of future satellites of the *CBERS* series and prepare proposals of technical requirements for such sensors.

The activities outlined above shall be implemented through specific cooperation projects.

ARTICLE IX

The Parties agree to make their best effort to facilitate the entry and exit of equipment and materials from the other Party necessary for the implementation of activities under this Complementary Protocol.

ARTICLE X

Subject to its laws and regulations, each Party shall facilitate, on a reciprocal basis, entry documentation for the other Party's nationals to enter and exit its national territory in order to carry out activities within the scope of this Complementary Protocol.

ARTICLE XI

The Parties should establish special arrangements concerning the issue of intellectual property, taking into account national laws and regulations of each country and international rules adopted by both Parties.

ARTICLE XII

Disputes concerning the interpretation or application of this Complementary Protocol shall be settled by mutual consultations between the Parties through diplomatic channels.

ARTICLE XIII

This Complementary Protocol shall enter into force upon Signature. This Complementary Protocol shall remain in force for five consecutive years. It shall be automatically renewed for equal and successive periods of five years, unless either of The

Parties notifies the other Party, through diplomatic channels, with a minimum of six months prior notice, of its intention to terminate this Protocol. Unless otherwise agreed between the Parties, the denouncement notice shall not affect on-going cooperation projects.

This Complementary Protocol may be amended by written agreement between the Parties.

Done in Brasilia on November 12, 2004, in duplicates, in the Portuguese, Chinese and English languages, all three texts being equally authentic. In case of any difference of interpretation, the English text shall prevail.

FOR THE GOVERNMENT OF THE FEDERATIVE REPUBLIC OF BRAZIL

FOR THE GOVERNMENT OF THE PEOPLE'S REPUBLIC OF CHINA

CBERS Data Policy

1. INTRODUCTION

This document defines the data policy directives for the *CBERS* Program, which includes provisions for reception, processing and dissemination of *CBERS* imagery to countries other than Brazil and China.

2. GENERAL CONSIDERATIONS

The downlink data is open to any country or organization and is based on the conception that *CBERS* imagery will be distributed by licensed representatives who operates an *application system infrastructure* that performs data reception and processing. In this document, the term *"ground station"* stands for *application system infrastructure.*

The data downlink for *CBERS* will be carried out through a ground station. International ground stations will not have access to the on-board data recorder (OBDR), which will be operated exclusively by CRESDA and INPE. Each ground station receives the image raw data and process it into image products, which will then be distributed to users. The licensing of *CBERS* data downlinks is based on fees which are charged in a per-minute basis.

China and Brazil may, in a few special cases, upon mutual consultation, decide on the transfer of data free of charge.

The revenues resulting from the distribution of *CBERS* data will be equally shared between China and Brazil, with accounting adjustments taking place every six (6) months.

3. THE BRAZILIAN AND CHINESE GROUND STATIONS

The ground stations operated by INPE in Brazil and by CRESDA in China have unlimited access to all data collected

within their footprint. The policy for distribution of data collected by those ground stations will be defined by each operator.

INPE and CRESDA will jointly agree on an international price list for *CBERS* images. Distribution of *CBERS* images to third parties will be done solely on the basis of the international price list, as agreed by China and Brazil, except in cases for which Brazil and China, upon mutual consultation, decide on the transfer of *CBERS* data free of charge.

4. USE OF THE ON-BOARD DATA RECORDER

Usage of the OBDR will be subject to the following guidelines:

a) The number of hours monthly available for the OBDR will be established periodically by the engineering teams of *CBERS*. INPE and CRESDA will equally share the available time, i.e, 50% to CRESDA and 50% to INPE, on a non-cumulative basis, such that unused hours in a month cannot be accumulated for subsequent periods. In special situations, the usage of time could be changed after mutual consultation.

b) Considering the lifetime and the reliability of the OBDR, usage of OBDR should be kept to a minimum level jointly specified by CRESDA, INPE, CAST and CLTC.

c) It is suggested that the OBDR should mainly be used in emergency situations and for demonstration and test purposes.

Images stored in the OBDR are downloaded at the Brazilian and Chinese ground stations. INPE and CRESDA will distribute these data according to the agreed international price list for *CBERS* images, except in special cases for which Brazil and China, upon mutual consultation, decide on the transfer of *CBERS* data free of charge.

5. DEVELOPMENT OF APPLICATION SYSTEM INFRASTRUCTURE

INPE and CRESDA shall agree on the policy for the construction or update of the receiving and processing systems to be installed at all, international, licensed ground stations, which will be defined in a specific document.

Information regarding data downlink, i.e., satellite to ground station interface, is regarded as intellectual property of

INPE and CRESDA and will not be disclosed to third parties unless agreed in writing by the parties. INPE and CRESDA will encourage Brazilian and Chinese companies to act as providers of the *CBERS* application system infrastructure.

INPE and CRESDA shall agree on a common specification for the *CBERS* application system infrastructure, hardware and software, and on a work breakdown structure which defines which component will be built by each party.

In the case of the Brazilian and Chinese ground stations, the parties shall agree on a joint specification. Each party is then free to decide on his development strategy that will preferably be done only by its national companies. The parties agree that any component of the ground station that cannot be built by its local industry will first be offered to the other partner, before any contracts are placed in the international market.

6. LICENSING POLICY FOR INTERNATIONAL GROUND STATIONS

International ground stations will be licensed according to the following guidelines:

a) *CBERS* data reception, processing and distribution to other countries will be carried out by licensed representatives jointly appointed by CRESDA and INPE.

b) The licensed representative will commercialize *CBERS* data downlink to ground stations based on a annual fixed basis, based on a fee determined by INPE and CRESDA. The annual fee will be determined by the conditions of the ground stations, including geographical location and antenna footprint.

c) The *CBERS* reception and production systems will be provided by Brazilian and Chinese companies, according to the provisions set on item 5.

d) The agreements for data reception are limited by technical capabilities of the satellite.

e) The data downlink to the ground station will have priority over use of the on-board data recorder, except in the case of emergency situations, as determined by INPE and CRESDA.

f) When requested by INPE or CRESDA, the licensed representative shall provide a copy of collected *CBERS* raw data.

g) Within the valid period of the license, INPE and CRESDA will provide technical support to the licensed representative, according to the provisions set in the licensing agreement.

h) The licensed representatives shall provide their customers satellite, receiving and processing parameters such as: satellite ephemeris, calibration data of the payloads, orbit holding information, satellite attitude control status data, satellite attitude control performance data and sensor performance attenuation information.

i) The licensed representatives shall hold annual meetings with their customers and promote the diffusion of any news or decisions about *CBERS*.

7. Product Distribution Policy

The commercial agreement between licensed representatives and distributors shall include the following:

a) The right of receiving, processing and distributing *CBERS* data shall be granted to the distributor by the licensed representative.

b) For the distributed products, definition, name, content, processing level, media, browsing and search mode, ordering procedure and after-service shall conform to a standard product format, specified in a specific document by INPE and CRESDA.

c) An authorization for a *CBERS* distributor will be granted only after the acceptance by CRESDA and INPE of a sample of its products.

d) INPE and CRESDA will set up, maintain and run a central catalog on *CBERS* data, including the metadata, browse data and related updated technological data.

e) All the browse images of *CBERS* data collected by the distributor must be sent to the central catalog at least once every month.

f) Each distributor could set its native price list independently for distribution solely within its respective national market. Images distributed within the distributor's national market may not be exported abroad.

g) When distributing abroad, the distributor must refer to the international price list set by INPE and CRESDA.

h) The distributor must provide its domestic price list, i.e., the price for its native users, to INPE and CRESDA.

i) The distributor shall provide an report of the data distribution every six months to INPE and CRESDA.

j) The distributor must provide and update the following documents for the users: (a) *CBERS* users' handbook; (b) standard PATH/ROW map; (c) *CBERS* orbit forecast; (d) application demonstration CD.